[SQL 2023] 12 <Access control>

*** ROLE**

Roles

[SQL 2023] 4.42.3 Roles

SQL makes no distinction between **users** and **groups** of users.

A **role** represents both.

A role may be a **member** of another role.

```
-- CREATE ROLE
--
CREATE ROLE yui
CREATE ROLE yui
  WITH ADMIN CURRENT_ROLE     -- as ADMIN, granted by
current role
  WITH ADMIN CURRENT_USER     -- same


-- DROP ROLE
DROP ROLE yui
DROP ROLE yui
  RESTRICT                -- Prevent dropping role if
there are associated object. Default
  CASCADE                 -- Force drop.



-- GRANT ROLE
--
-- Role membership
GRANT arole TO yui            -- add yui to arole
GRANT arole, brole            -- multiple
  TO yui, usera
GRANT arole
  TO yui
  WITH ADMIN OPTION           -- ADMIN of CREATE ROLE

-- CURRENT_ROLE: special builtin role
GRANT CURRENT_ROLE TO yui
GRANT someRole TO CURRENT_ROLE


-- REVOKE ROLE
--
REVOKE arole FROM yui
REVOKE arole, brole       -- multiple
  FROM yui, auser
REVOKE ADMIN OPTION FOR yui
```

*** PRIVILEGE**

[SQL 2023] 12.2 <grant privilege statement>

Privileges
Inheritance
[SQL 2023] 4.42.2 Privileges
A role is assigned **privileges**.
A role by **inherits** the privileges by membership by default.
A role may optionally revoke inherited all or single privileges.

Types of targets:

- TABLE
- TABLE Columns
- DOMAIN
- COLLATION
- CHARACTER SET
- TRANSLATION
- TYPE
- SEQUENCE

Types of privileges:

SELECT	allow SELECT of any column from table
SELECT (col1, col2)	allow SELECT from specific column only, if whole table is REVOKEd
INSERT	allow INSERT any column in table
INSERT (col1, col2)	allow INSERT into specific columns only, if whole table is REVOKEd
UPDATE	
UPDATE (col1, col2)	
REFERENCES	foreign key
USAGE	
TRIGGER	create a trigger
EXECUTE	call a function
UNDER	
ALL PRIVILEGES	

GOTCHA: column-specific privileges < table privileges
Column specific are effective only if table privileges are revoked

e.g.

```
GRANT SELECT ON mytable TO user1;
GRANT SELECT (x) ON mytable TO user1;      -- BAD

GRANT SELECT (x) ON mytable TO user2;      -- GOOD

-- BAD: user 1
SELECT * FROM mytable;                     -- SUCCESS
SELECT (x) FROM mytable;                   -- SUCCESS

-- BAD: user 2
SELECT * FROM mytable;                     -- ERROR
SELECT (x) FROM mytable;                   -- SUCCESS
```

```
-- Privileges
--
GRANT INSERT ON mytable TO yui
GRANT INSERT, UPDATE ON mytable    -- multiple privileges
  TO usera, userb                  -- multiple userrs
GRANT
  INSERT,
  INSERT (columna, columnb),
  UPDATE,
  UPDATE (columna, columnb),
  SELECT
  SELECT (columna, columnb),
  USAGE,
  ...
  ON TABLE mytable
  TO yui

-- OPTIONS
GRANT ... TO yui
  WITH GRANT OPTION                -- allow yui to GRANT
same privileges to other roles
  WITH HIERARCHY OPTION


REVOKE INSERT ON mytable FROM yui
...
```